

NIKOLA®

INFORMATION FOR FIRST AND SECOND RESPONDERS

EMERGENCY RESPONSE GUIDE



NIKOLA

TRE BEV

BATTERY ELECTRIC



VERSION: 003



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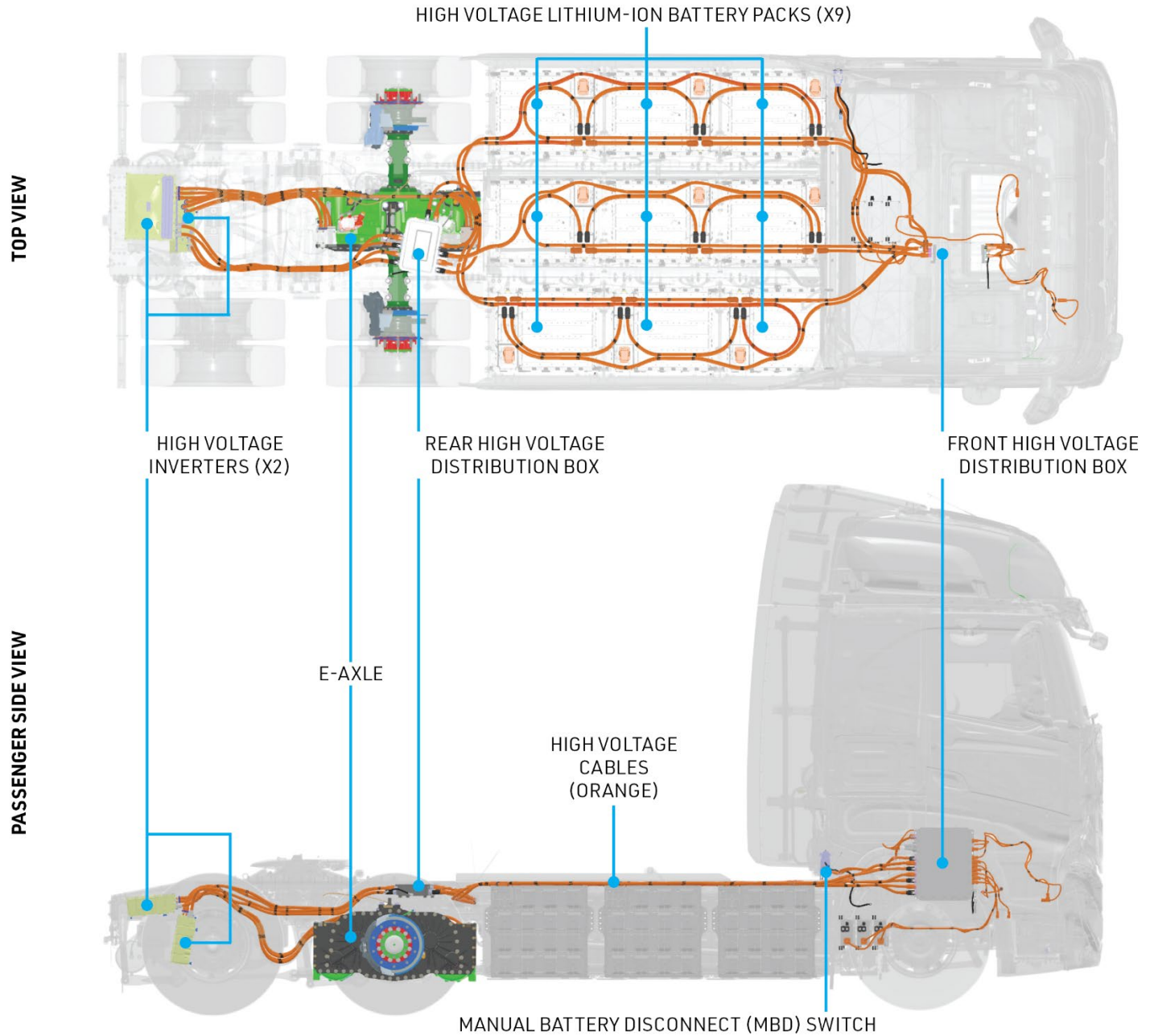
High Voltage (HV) System Diagram



WARNING: All high voltage (HV) wires and harnesses are wrapped in orange insulation. NEVER cut these sections.

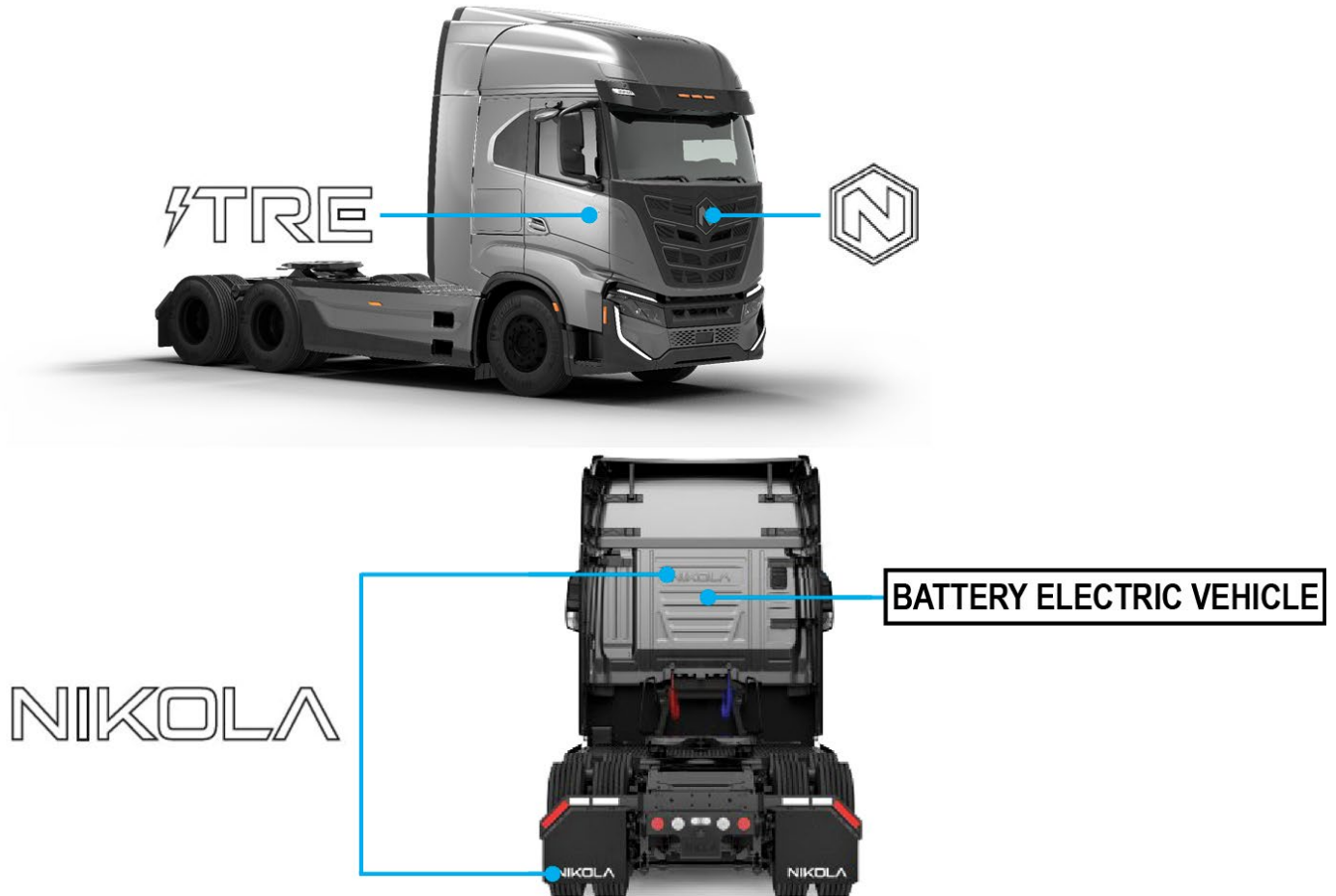


Do not try to touch without proper PPE (HV insulated gloves and boots, safety glasses, etc.). Failure to follow these instructions may result in the risk of serious injury or death.





1. Identification / Recognition



Vehicle Identification Number (VIN):

All Nikola manufactured vehicles are equipped with a HV system and can be identified as a Nikola vehicle by the first three characters of the VIN which will always be "1N9."

Furthermore, the 6th character of the VIN will either be a "B" to indicate battery electric vehicle or an "H" to indicate hydrogen fuel cell electric vehicle.

The VIN is found on the driver side A-pillar and stamped on the frame railing inside driver side wheel well.



2. Immobilization / Stabilization / Lifting



WARNING: Do not touch, cut, or open high-voltage components and/or high-voltage batteries.



Always wear the full firefighting PPE and proper insulated electrical PPE (safety glasses, gloves rated for at least 1,000 V, HV insulated safety shoes, etc.).

Remove all metallic jewelry including rings, chains, and watches.

Failure to follow these instructions may result in the risk of serious injury or death.

A. Chock the trailer and truck wheels.



B. Activate the parking brakes by lifting up on the right side of both the yellow and red switches.



C. If the vehicle is on (START/STOP button LED is green), and time allows, turn it off by pressing the START/STOP button. Verify the START/STOP button LED is off.



D. Turn the MBD switch counterclockwise (CCW) to off.

MBD ON

MBD OFF



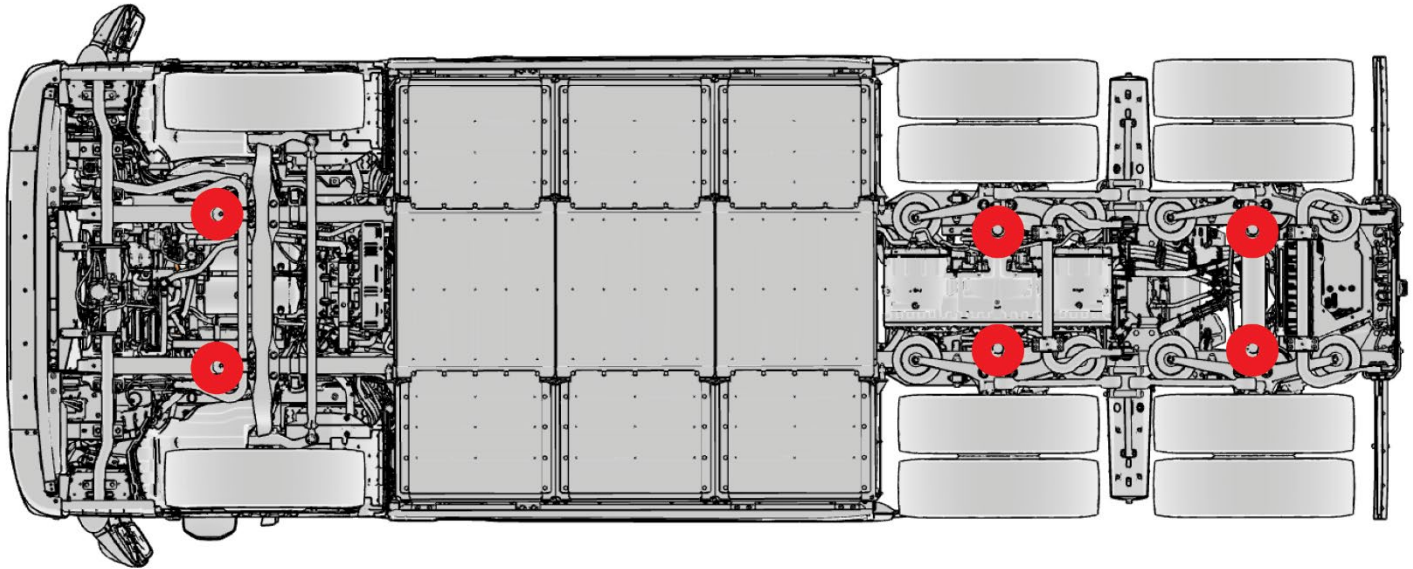


Lifting Points:

The Vehicle can be lifted/jacked at the points along each axle shown in the figure below.



NOTICE: Never stabilize or lift the vehicle directly from the battery pack or anywhere along the high voltage system.





3. Disable Direct Hazards / Safety Regulations

Disabling High Voltage (HV):



WARNING: It is possible for the HV system to keep significant levels of voltage for a short duration after the system has been deactivated. Allow 5 minutes for voltage to de-energize before interacting directly with HV components.

Do not touch, cut, or open high-voltage components and/or high-voltage batteries.

Always wear full firefighting and proper insulated electrical PPE (safety glasses, gloves rated for at least 1,000V, HV insulated safety shoes, etc.). Remove all metallic jewelry including rings, chains, and watches.

Failure to follow these instructions may result in the risk of serious injury or death.



- Fold the passenger side aero fairing to access manual battery disconnect (MBD) switch
- Locate the MBD switch
- Turn the MBD switch counterclockwise (CCW) to off
- Wait a minimum of **5 minutes** for the high voltage system to de-energize

MBD ON



MBD OFF





4. Access to the Occupants

Windows:



WARNING: Electrical and mechanical components (seats, doors, steering wheel, etc.) may be compromised after a collision.



If extrication is needed, do not touch, cut, or open high-voltage components and/or high-voltage batteries.

Failure to follow these instructions may result in the risk of serious injury or death.

The side windows and sunroof glass on the Tre BEV are made of tempered glass.
The windshield is made of laminated safety glass.



1. Tempered Glass
2. Laminated Safety Glass



5. Stored Energy / Liquids / Gases / Solids



WARNING: If coolant and/or refrigerant escapes from the system, there is a risk of thermal reaction within the high voltage battery system.



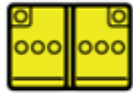
If available, use thermal imaging to monitor temperature of the high voltage battery for thermal reaction.



Failure to follow these instructions may result in the risk of serious injury or death.



800V



12V and 24V



Coolant (R1234yf)



6. In Case of Fire

If the fire is detected early enough when it is small and contained in a single area, spray enough water to quickly extinguish the flame.

Do not store a vehicle holding a burned Li-Ion HV battery in or within 50 feet of a structure or other vehicle until the battery can be discharged.



WARNING: Follow all safety guidelines and procedures before entering the emergency scene.



All personnel should wear and use full firefighting PPE and SCBA as needed at all vehicle fires.

Use copious amount of water to extinguish battery fires.

Failure to follow these instructions may result in the risk of serious injury or death.

USE COPIOUS AMOUNTS OF WATER TO FIGHT A HIGH VOLTAGE BATTERY FIRE.

If the battery catches fire, is exposed to high heat, or is generating heat or gases, use large amounts of water to cool the battery. It can take over 2,600 gallons (9,842 liters) of water (applied directly to the battery) to effectively cool and extinguish a high voltage battery fire.

If a Lithium Ion (Li-Ion) HV battery is involved in a fire, there is a possibility that it could reignite after being extinguished. If available, use thermal imaging to watch the battery. Re-apply water as necessary to cool the battery packs. Battery packs must be kept below 248°F (120°C). Alternatively, burying the battery packs in sufficient amounts of sand to cover the entire vehicle mid-section may achieve a similar effect and can be used to prevent re-ignition of battery packs.



Do not store a vehicle containing a damaged or burned Li-Ion HV battery in or within 50 feet of a structure or other vehicle until the battery can be discharged or has been removed from the vehicle and disposed.



7. In Case of Submersion



DANGER: Damaged electric vehicles submerged in water present a potential high voltage electrical shock hazard. Exercise caution and wear proper PPE safety gear, including, but not limited to, high voltage safety gloves and boots.

Failure to follow these instructions will result in the risk of serious injury or death.

Battery electric vehicles that have been submerged in water should be handled with greater caution due to the potential risk of a high voltage electrical battery fire. First responders should be prepared to respond to a potential fire risk. Pull the vehicle out of the water and allow the water to drain out of the vehicle's high voltage battery packs. Follow section 3 of this document, **Disable Direct Hazards / Safety Regulations**, once the vehicle is out of the water.



8. Towing / Transportation / Storage

Towing

The vehicle can be pushed or pulled a short distance to clear it from the roadway. Once cleared, refer to the towing procedures to tow the vehicle. If you can, place the vehicle in Neutral, and disengage the parking brake prior to pushing or pulling.



NOTICE: Damage to the E-axle. Pulling the vehicle with the E-axle wheels turning may cause significant damage to powertrain components. When pulling the vehicle with the E-axle wheels turning, DO NOT exceed a speed of 3 mph (4.8 km/h) for a short distance. Pulling the vehicle in this manner should be extremely limited and only used for repositioning the vehicle to safely hook with a proper procedure.

The E-axle in the vehicle can generate power if the wheels spin during towing which could lead to overheating and cause significant damage. The vehicle can be towed from the rear (reverse tow or 5th wheel lift method), the front (with front hook), or with a lowboy trailer. Always transport the vehicle with the E-axle off the ground and restrict the wheels from spinning. The vehicle must remain in Neutral during towing.



Front Hook

1. Put the vehicle in the accessory mode.
2. Set the parking brake **(1)** and install wheel chocks along the tag axle.
3. Locate the ECAS remote control on the left side of the driver's seat.





4. Press the consent button **(1)** on the ECAS remote control.
5. Press the lowering button **(2)** to set ECAS to the lowest position.



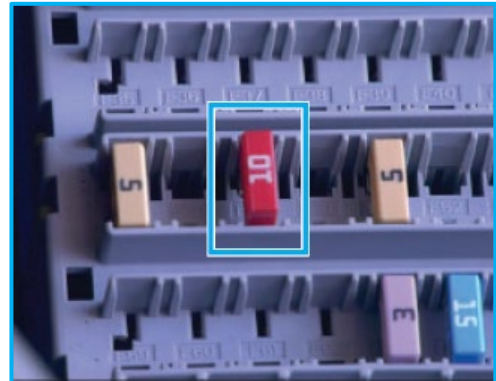
6. Turn vehicle off in the cab.
7. Set the MBD switch to the OFF position.



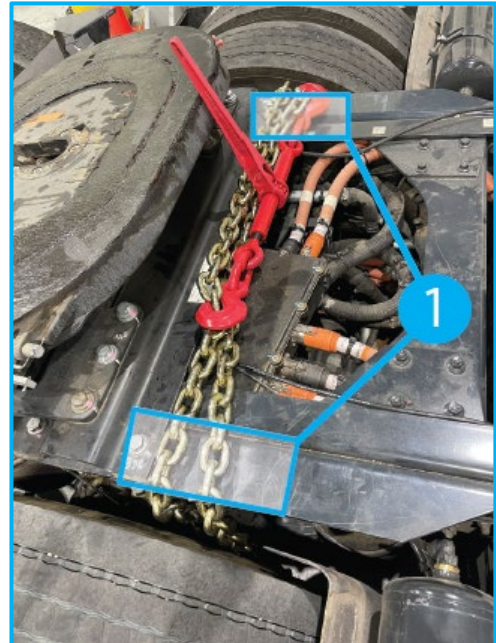


8. Deactivate the ECAS by switching off the vehicle power and removing 10A fuse from position **F49** in the passengers side dashboard fuse panel.

Note: This will deactivate ECAS and will not compensate for any ride height changes during the towing procedure.

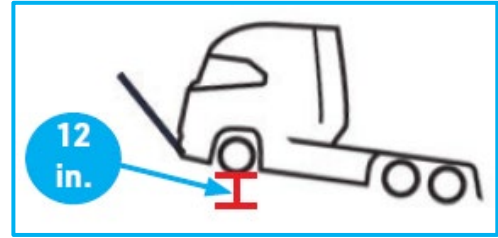


9. Chain the E-Axle to the frame.
10. Place two blocks **(1)** between the frame rails and chains to keep the chains from damaging the frame rails.

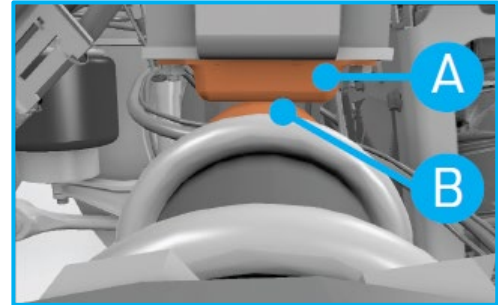




11. Lift the vehicle up from the front axle such that the tires are 12 inches above the ground.

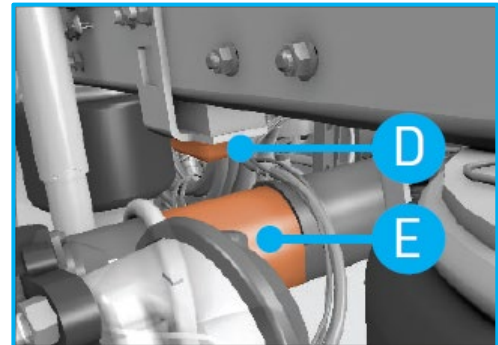
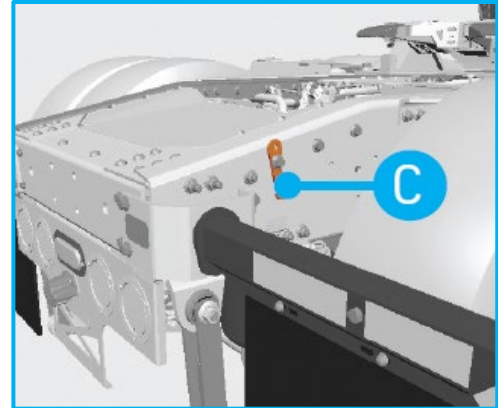


Note: This tilts the truck and loads the tag axle more and compresses the tag axle air suspension. The tag axle **(B)** will contact the bump stops **(A)** if the vehicle's front end is lifted while the trailer is still hitched.



12. Locate the manual air inlet port **(C)** on the tag axle.
13. Use the manual air inlet port and an external air source to pressurize the tag axle air springs to lift the frame above the tag axle so that there is **2 in. (52 mm)** of clearance between the bottom of the rubber portion of the bump stop **(D)** and to the top of the tag axle **(E)**.

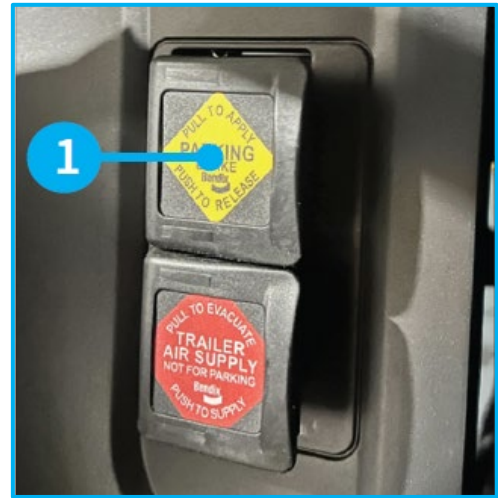
Note: This allows for tag axle jounce travel while towing with the trailer.



14. Disconnect the external air source line from the manual air inlet port.



15. Disengage the parking brake **(1)**.



16. Remove the wheel chocks from the tag axle.
Note: The vehicle is now ready for towing.
17. Place all tow kit contents back into the tow kit package, including the previously removed ECAS fuse.

PROCEDURE IS COMPLETE



Rear Hook (no trailer) with Reverse Kingpin / 5th Wheel Lift

1. Put the vehicle in the accessory mode.
2. Set the parking brake **(1)** and install wheel chocks along the tag axle.



3. Secure the steering wheel into a non-moving position.



4. Locate the ECAS remote control on the left side of the driver's seat.





5. Press the consent button **(1)** on the ECAS remote control.
6. Press the lowering button **(2)** to set ECAS to the lowest position.



7. Turn vehicle off in the cab.

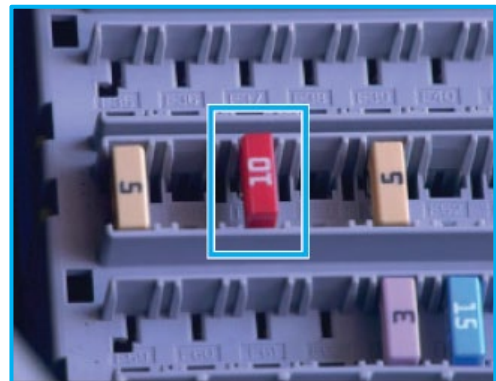


8. Set the MBD switch to the OFF position.



9. Deactivate the ECAS by switching off the vehicle power and removing 10A fuse from position **F49** in the passengers side dashboard fuse panel.

Note: This will deactivate ECAS and will not compensate for any ride height changes during the towing procedure.

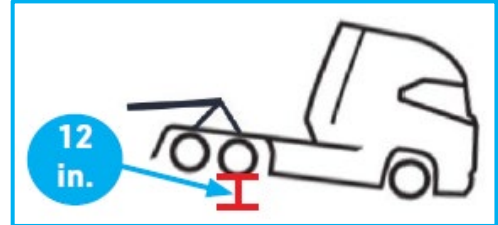




10. Chain the E-axle and tag axle in a triangle formation, avoiding contact with the air tanks.



11. Lift the vehicle up from the rear axles such that the tires are 12 inches above the ground.



12. Disengage the parking brake **(1)**.



13. Remove the wheel chocks from the tag axle.

Note: The vehicle is now ready for towing.

14. Place all tow kit contents back into the tow kit package, including the previously removed ECAS fuse.

PROCEDURE IS COMPLETE



Transportation

The high voltage system may be compromised because of a collision. Before transporting, it is important to assume these components are energized. Always follow high voltage system safety precautions and avoid transporting until the HV system has been properly deenergized. Failure to do so may result in serious injury.

Storage



WARNING: The potential for battery re-ignition exists after a crash or fire incident. Vehicle should be stored outside at a safe distance (50 ft minimum) from other vehicles and structures.



Failure to follow these instructions may result in the risk of serious injury or death.

Damaged battery electric vehicles should be kept in an open area (instead of in a garage or other type of enclosed storage facility) until the batteries have been properly discharged.

For further instructions on storing and/or scrapping high voltage battery system components, contact Nikola Service at **888.429.9216, Option 1**.

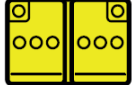


9. Important Additional Information

First responders and training officers with questions may contact Nikola Service at **888.429.9216, Option 1**.



10. Explanation Pictograms Used

	Acute Toxicity
	Battery Electric Vehicle
	Caution / Danger
	Corrosive
	Explosive
	Flammable
	Gases under pressure
	Hazards to the Human Health
	High Voltage Battery
	High Voltage Warning
	Infrared Imaging
	Low Voltage Battery